DRAFT STANDARD OPERATING PROCEDURE

Single Seedling Plot Test (SSPT)

Sugarcane Breeding and Germplasm Conservation Program

SOP No.	SCB-BR-SOP-05	**Revision**	0
Effective date	_____	**Supersedes**	New
Prepared by	_____	**Reviewed by**	_____
Approved by	_____	**Controlled copy**	Yes / No

Status: Draft technical SOP. Field dimensions, thresholds, chemical treatments, quarantine steps, and local compliance requirements must be validated and approved by the implementing institution before adoption.

1. Purpose

To evaluate large populations of genetically unique sugarcane seedlings from controlled crosses and select superior individuals for advancement to the Row Test or first clonal evaluation stage while preserving complete pedigree and field traceability.

2. Scope

Applies from hardened true-seed seedlings ready for field transplanting through plant-cane observation, selection, tagging, harvest, and advancement. In the Sugar Regulatory Administration breeding pipeline, SSPT is a recognized early selection stage followed by Row Test.

3. Selection philosophy

A single seedling is genetically unique and normally cannot be replicated at this stage. Therefore, selection precision is lower than in later clonal trials. Use strong negative selection for clear defects and disease, but avoid excessively severe early selection based only on one environmentally sensitive trait. Where family sizes permit, combine between-family and within-family selection.

4. Required inputs and records

- Cross register with verified female x male pedigree.
- Seedbox/tray and family identity records.
- Field map generated before transplanting.
- Commercial check/control varieties repeated through the field.
- Seedling tags or position-based identification system.
- Selection tags, field notebook/mobile data system, refractometer, measuring tools, and disease scoring guides.

5. Traceability rule

Maintain an unbroken chain: Cross number -> seed lot -> seed box/tray -> family -> field row/position -> selected stool -> provisional clone number -> Row Test position. A selected stool must never be detached from its pedigree record.

6. Field layout options

- Family-grouped layout: seedlings from the same cross planted in contiguous family groups, with repeated checks between groups or blocks.
- Augmented design: unreplicated seedling entries with replicated standard checks distributed across the field to help account for spatial variation.
- Whichever design is used, record row, plant position, family/cross, and check positions before transplanting.

7. Transplanting procedure

1. Select healthy, adequately hardened seedlings with correct family labels. Reject severely stunted, diseased, mixed, or unidentified trays before field movement.
2. Move one family at a time from nursery to field to minimize accidental mixing.
3. Transplant according to the preprinted field map; maintain the program's validated plant spacing and row spacing.
4. Place repeated commercial checks systematically across blocks, not only around the perimeter.
5. Record missing plants and any substitutions; never fill a dead seedling position with an unidentified seedling.
6. Audit field position versus family label after establishment and before labels become obscured by canopy growth.

8. Observations during establishment

- Survival and establishment vigor.
- Obvious disease symptoms or severe susceptibility.
- Tillering and stool establishment.
- Gross abnormality, off-type status, or contamination.
- Field problems such as waterlogging, drought patches, or herbicide injury that could bias later selection.

9. Selection observations at maturity

Use a combination of traits; do not select on Brix alone.

Trait group	Recommended observations	Breeder interpretation
Yield components	Tillers/millable stalks; stalk height; diameter; stalk/stool weight where feasible	Seek balanced yield potential, not a single oversized stalk
Juice quality	Hand-refractometer Brix; top/middle/bottom Brix if used; maturity	Compare with checks at similar age
Agronomic type	Erectness; lodging; stalk uniformity; trashing; suckering; flowering	Penalize traits that reduce harvestability or stability unless strategically valuable
Health	Smut and other locally important disease symptoms; insect injury	Strong rejection criterion for major susceptibility
Ratoon potential	Where SSPT is ratooned: regrowth, stand, vigor	Useful confirmation before advancement when program design allows

10. Family-based selection

When families contain enough seedlings, first rank family performance using family means or the proportion of acceptable stools, then make within-family selections among the better families. Retain strategic diversity: a smaller number of exceptional seedlings from a genetically valuable family can be justified even if the family mean is modest, provided the selection decision is documented.

11. Advancement procedure

7. Tag candidate stools in the field before harvest using a unique temporary selection ID linked to row/position and cross.
8. Recheck each tagged stool on a separate visit when feasible to reduce one-day visual bias.
9. Assign provisional clone numbers only after identity and selection status are verified.
10. Cut each selected stool separately. Place the label with the stalks immediately; do not create unlabelled piles.
11. Transport selections in physically separated, clearly labeled bundles.
12. Plant the Row Test / first clonal test according to a prepared map, then reconcile selected-stool count against planted clone count.
13. Archive both selected and rejected population statistics by family to inform future parent and cross decisions.

12. Quality-control checkpoints

- 100% of selected stools have known field position and cross/pedigree.
- Checks are distributed through the field and recorded.
- Environmental problem areas are mapped and considered in selection decisions.
- No selected stalk is moved without a physical label.
- Selection intensity is calculated overall and by family.
- Advancement list reconciles with the next-stage planting list.

13. SSPT selection sheet

Row/Pos.	Cross	Vigor	Stalk no.	Diameter	Brix	Health	Lodging/ flower	Decision	Clone no.

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References and technical basis

Sugar Regulatory Administration - research portfolio: Pollination/Sowing/Seedling Care and SSPT projects - Web source

SRA 2016 Annual Report - example SSPT advancement to Row Test - Web source

Sugar Research Australia - family/progeny assessment approach - Web source